№ 76.

$$\xi \sim p(x,\theta) = \begin{cases} \dfrac{\theta-1}{x^{\theta}}, & x \geq 1 \\ 0, & x < 1 \end{cases}, \text{ где } \theta > 1 \quad \exists \overrightarrow{X_n}$$

a) ОМП

ф-я правдоподобия:

$$L(\theta) = \prod_{i=1}^{n} p(x_i, \theta) = \frac{(\theta-1)^n}{\left(\prod_{i=1}^{n} x_i\right)^{\theta}} \left\{ \min_{i=1,\dots,n} x_i \geq 1 \right\}$$

$$(\ln L)'_{\theta} = \left(n\ln(\theta-1) - \theta\left(\sum_{i=1}^{n} \ln x_i\right)\right)'_{\theta} = \frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i$$

$$\frac{n}{\theta-1} - \sum_{i=1}^{n} \ln x_i = 0 \Rightarrow \underset{\text{ОМП}}{\tilde{\theta} = 1 + \frac{n}{\sum_{i=1}^{n} \ln x_i} = 1 + (\overline{\ln x})^{-1}}$$

$$(\ln L)''_{\theta\theta} = \frac{-n}{(\theta-1)^2} < 0 \Rightarrow \text{это максимум.}$$

b) ОМП

Докажем, что ПВМ – сильно регул.:

1) явл. регулярной

- $p(x,\theta)$ – непр. дифф. по θ на $(1;+\infty)$
- $\int_{1}^{+\infty} \frac{\partial}{\partial\theta} p(x,\theta)\,dx \overset{?}{=} 0$

$$p'_{\theta} = \left(\frac{\theta-1}{x^{\theta}}\right)'_{\theta} = \frac{\ln x - \theta\ln x + 1}{x^{\theta}}$$

на $(1;+\infty)$

Torga $\int\limits_1^{+\infty} x^{-\theta}\left(\ln x(1-\theta)+1\right)dx = \frac{x^{-\theta+1}}{1-\theta}\Big|_1^{+\infty} + \int\limits_1^{+\infty} \ln x \cdot d\left(x^{1-\theta}\right) =$

$$= \frac{1}{\theta-1} - \int\limits_1^{+\infty} x^{-\theta}dx = \frac{1}{\theta-1} - \frac{1}{\theta-1} = 0$$

$$I(\theta) = \int\limits_1^{+\infty} \frac{\theta-1}{x^{\theta}} \cdot \left(\left(\ln(\theta-1) - \theta\ln x\right)'_{\theta}\right)^2 dx =$$

$$= \int\limits_1^{+\infty} \frac{\theta-1}{x^{\theta}}\left(\frac{1}{\theta-1} - \ln x\right)^2 dx = \int\limits_1^{+\infty}\left(\frac{1}{x^{\theta}(\theta-1)} - 2\frac{\ln x}{x^{\theta}} + (\theta-1)\cdot\right.$$

$$\left.\cdot\frac{\ln^2 x}{x^{\theta}}\right)dx = \frac{1}{(\theta-1)^2} - 2\int\limits_1^{+\infty}\ln x\, d\left(-\frac{1}{(\theta-1)x^{\theta-1}}\right) +$$

$$+\int\limits_1^{+\infty}\frac{(\theta-1)\ln^2 x}{x^{\theta}}dx = \frac{1}{(\theta-1)^2} - 2\tilde{I} + \tilde{\tilde{I}}$$

$$\tilde{I} = \int\limits_1^{+\infty}\ln x\, d\left(\frac{-1}{(\theta-1)x^{\theta-1}}\right) = \underbrace{\ln x \cdot \frac{x^{1-\theta}}{1-\theta}\Big|_1^{+\infty}}_{\substack{\parallel \\ 0 \\ \text{m.k. } \ln 1}} - \int\limits_1^{+\infty}\frac{x^{-\theta}dx}{1-\theta} = \frac{-1}{(\theta-1)^2}$$

$$\tilde{\tilde{I}} = \int\limits_1^{+\infty}\frac{(\theta-1)\ln^2 x}{x^{\theta}}dx = \int\limits_1^{+\infty}\ln^2 x\, d\left(-x^{1-\theta}\right) = -\left(\underbrace{\ln^2 x \cdot x^{1-\theta}\Big|_1^{+\infty}}_{\substack{\parallel \\ 0}} -\right.$$

$$\left.-\int\limits_1^{+\infty} 2\cdot\frac{\ln x}{x}\cdot x^{1-\theta}dx\right) = 2\int\limits_1^{+\infty}\ln x \cdot x^{-\theta}dx = 2\int\limits_1^{+\infty}\ln x\, d\left(\frac{x^{1-\theta}}{1-\theta}\right) =$$

$$= 2\left(\underbrace{\frac{x^{1-\theta}}{1-\theta}\ln x\Big|_1^{+\infty}}_{\substack{\parallel \\ 0}} - \int\limits_1^{+\infty}\frac{x^{-\theta}dx}{1-\theta}\right) = \frac{-2}{(\theta-1)^2}$$

$$\Rightarrow I(\theta) = \frac{1}{(\theta-1)^2} + 2\cdot\frac{1}{(\theta-1)^2} - \frac{2}{(\theta-1)^2} = \frac{1}{(\theta-1)^2}$$

непр. и >0 на $(1;+\infty)$

2) $p(x,\theta)$ – $k \geq 2$ раз непр. дифф. по θ на $(1;+\infty)$ (очевидно).

3) $\int\limits_1^{+\infty} \frac{\partial^l}{\partial\theta^l} p(x,\theta)dx \overset{?}{=} 0$ – верно для $l=1$.

Проверим $l=2$:

$$p''_{\theta\theta} = \ln x \cdot x^{-\theta}(\theta\ln x - \ln x - 2)$$

на $(1;+\infty)$

$$\int\limits_1^{+\infty}(\ln^2 x \cdot x^{-\theta}(\theta-1) - 2\ln x \cdot x^{-\theta})dx = 0$$

везде будет $\ln 1$.

$\Rightarrow$ ОМП состоят., асимпт. несмещ., асимпт. эффект. и асимпт. нормальн.

по опр. медиана: $F(X_{0,5}) = 0,5$.

т.е. $\int\limits_1^{X_{0,5}} \frac{\theta-1}{x^\theta}dx = \frac{-1}{X_{0,5}^{\theta-1}} + 1 = 0,5$

$$\Rightarrow X_{0,5} = 2^{\frac{1}{\theta-1}}$$

Тогда $g(\tilde\theta) = \sqrt[\tilde\theta-1]{2}$

Имеем: $\sqrt{n}\,\frac{g(\tilde\theta)-g(\theta)}{\sqrt{\vec\nabla^T g(\tilde\theta) I^{-1}(\tilde\theta)\vec\nabla g(\tilde\theta)}} \rightsquigarrow N(0,1)$

$$\vec{\nabla} g(\tilde{\theta}) = 2^{\frac{1}{\tilde{\theta}-1}} \cdot \ln 2 \cdot \frac{-1}{(\tilde{\theta}-1)^2}$$

$$\sqrt{n}\,\frac{(g(\tilde{\theta})-g(\theta))(\tilde{\theta}-1)}{2^{\frac{1}{\tilde{\theta}-1}} \cdot \ln 2} \rightsquigarrow N(0,1)$$

т.о.

$$P\left(g(\tilde{\theta}) - \frac{1{,}96 \cdot \ln 2 \cdot 2^{\frac{1}{\tilde{\theta}-1}}}{\sqrt{n}\,(\tilde{\theta}-1)} < x_{0,5} < g(\tilde{\theta}) + \frac{1{,}96 \cdot \ln 2 \cdot 2^{\frac{1}{\tilde{\theta}-1}}}{\sqrt{n}\,(\tilde{\theta}-1)}\right) = \beta.$$

асимптотический доверительный интервал (ОМП) для медианы

c) ОМП

$$\frac{\sqrt{n}\,(\tilde{\theta}-\theta)}{\tilde{\theta}-1} \rightsquigarrow N(0,1)$$

$$-1{,}96 < \sqrt{n} \cdot \overline{\ln x}\left(1 + \frac{1}{\overline{\ln x}} - \theta\right) < 1{,}96$$

$$P\left(1 - \frac{1{,}96 - \sqrt{n}}{\sqrt{n}\,\overline{\ln x}} < \theta < 1 + \frac{1{,}96 + \sqrt{n}}{\sqrt{n}\,\overline{\ln x}}\right) = \beta$$

асимптотич. доверит. интервал (ОМП).